Name of Technology: **Nano silver screen-printable ink for flexible Electronics Applications"**

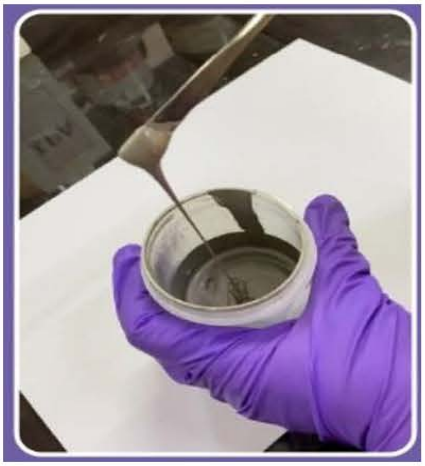
Printed electronics have gained interests from researchers in academia and industries. It is an emerging manufacturing technology to fabricate electronic and display appliances with printing technologies, which have been employed in conventional graphic art industries, at an ultra-low cost. The employment of conventional printing technologies to printed electronics might be traced back to the 1980.Since then, diverse electronic components have been demonstrated with various printing technologies. Even display applications such as thin film transistor liquid crystal, polymer light emitting diodes and flexible displays have attempted to adopt printing technologies as their manufacturing processes.

Printed commodities integrated with transistors, logics, batteries and electrochromic displays were also demonstrated with roll-to-roll based manufacturing equipment's. Among many possible applications with printing technologies, *Flexible Electronics have gained attention of industries because its market is believed to be wide open soon.*

C-MET has developed the low temperature (upto 150^{0}C depending on substrate) curable silver ink fir flexible electronics applications. Presently these pastes are imported from the handful of suppliers having worldwide monopoly. This paste consists of specially developed silver powder and dispersed in a suitable organic vehicle.

C-MET has filed an Indian patent application (Application No. 202211029539 dated 23.05.2022, publication date 29.12.2023) related to this technology.

Formulated Silver paste



Product specifications: Silver conductive paste (Batch:50gm)

Parameters	In-house developed Silver Paste	Commercial Paste (CI1001 make ECM, USA)
Viscosity (Cp)	8000-10000 Cp @30°C	7000-9000 CP @ 30° C.
Solids Content	60-70%	61%
Film Thickness	25 microns ± 5	25 microns
Sheet Resistance	0.01 – 0.07 Ohm/sq @ 25microns	< 0.015 Ohm/square @ 25 microns
Compatible Substrate	PET, PI, Paper, PVC	PET, PI, Paper
Curing Temperature	(60-150ºC)	110° C
Screen	77 T -120 T	120 T
Squeegee	75-80 Shore hardness	NA